



*Software Engineering (3) Course (SWE521): Agile  
Software Engineering Skills*

# **AGILE SOFTWARE ENGINEERING SKILLS**

## **LECTURE 01**

### **Introduction and Principles**

2024 / 2025

 Springer

Julian Michael Bass



# Outline

- 1. Introduction**
- 2. Agile Software Skills**
  1. People
  2. Product
  3. Process
  4. Advanced Skills
- 3. Engineering Software**
- 4. Tabby Cat Project**
- 5. Create Yourself a Livelihood**



# Overview– Introduction and Principles

- Software engineering is An engineering discipline with systematic methods.
- Software engineering framed around three core areas:
  - **People**: Working effectively in teams
  - **Product**: Building useful, secure software
  - **Process**: Following **repeatable** development cycles
- Case study used throughout: **Tabby Cat Project**
- **Notes**: *Introduce the chapter as both practical and strategic.*
  - *Emphasize the journey learners will take through team dynamics, technical know-how, and disciplined execution.*



# Why Agile Software Skills Matter?

- Software is everywhere—quality is critical
- Poor software leads to failure and risk (costs, usability, security)
- Agile skills = better teamwork, better software
- Early-career developers need practical tools to succeed
- Notes: relatable examples: a frustrating app, a failed online form, etc. Emphasize this isn't just theory—it's real-world preparation.



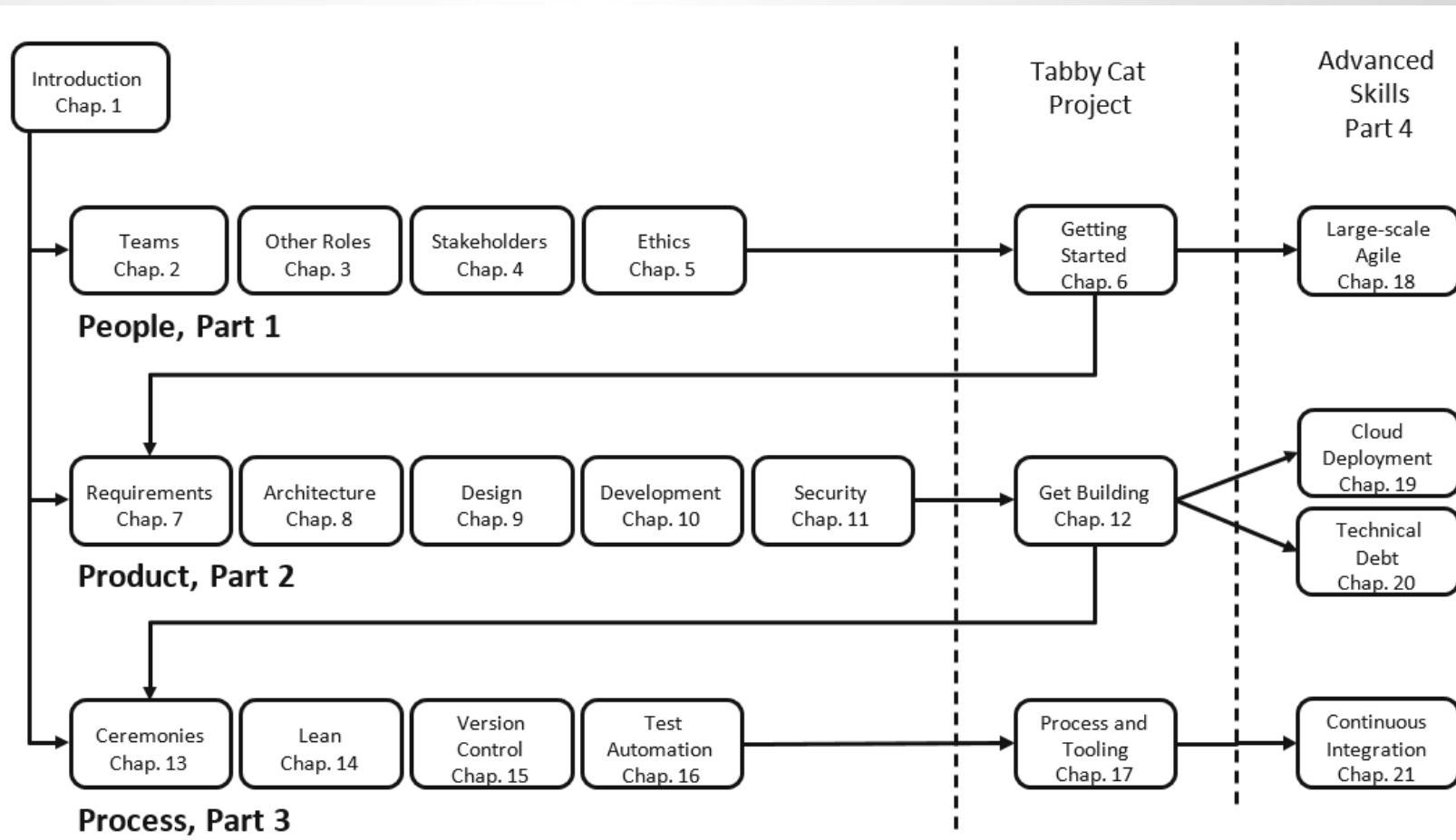
## Book Structure – People, Product, Process

- Part I: **People** – Teams, roles, ethics
- Part II: **Product** – Features, architecture, implementation
- Part III: **Process** – Ceremonies, testing, delivery
- Tabby Cat ties all parts together through a practical example

Notes: How learners move from team skills to building and deploying software.



# Book Structure





# The "People" Perspective

Collaboration is key: software is a team sport

- Covered chapters:
- Chap. 2 – Self-organizing teams
- Chap. 3 – Supporting roles
- Chap. 4 – Stakeholders and clients
- Chap. 5 – Ethics in tech practices

Note: Encourage stories of teamwork or conflict—help learners relate personally.



# The "Product" Perspective

- Build from requirements to features
- Features include UI, backend logic, and storage
- Delivered incrementally—small releases build toward the full product
- Requirements may evolve—discovery is part of the process
- Incremental delivery delivers parts of a system in stages

Notes: a real-world analogy like building a car feature-by-feature (e.g., GPS, then backup camera).



# Architecture, Design, Development & Security

- Chap. 8 – Architecture styles (layered, microservices)
- Chap. 9 – Object-oriented design patterns
- Chap. 10 – Implementation techniques
- Chap. 11 – Security strategies



# The "Process" Perspective

**Software process ensures consistency and quality**

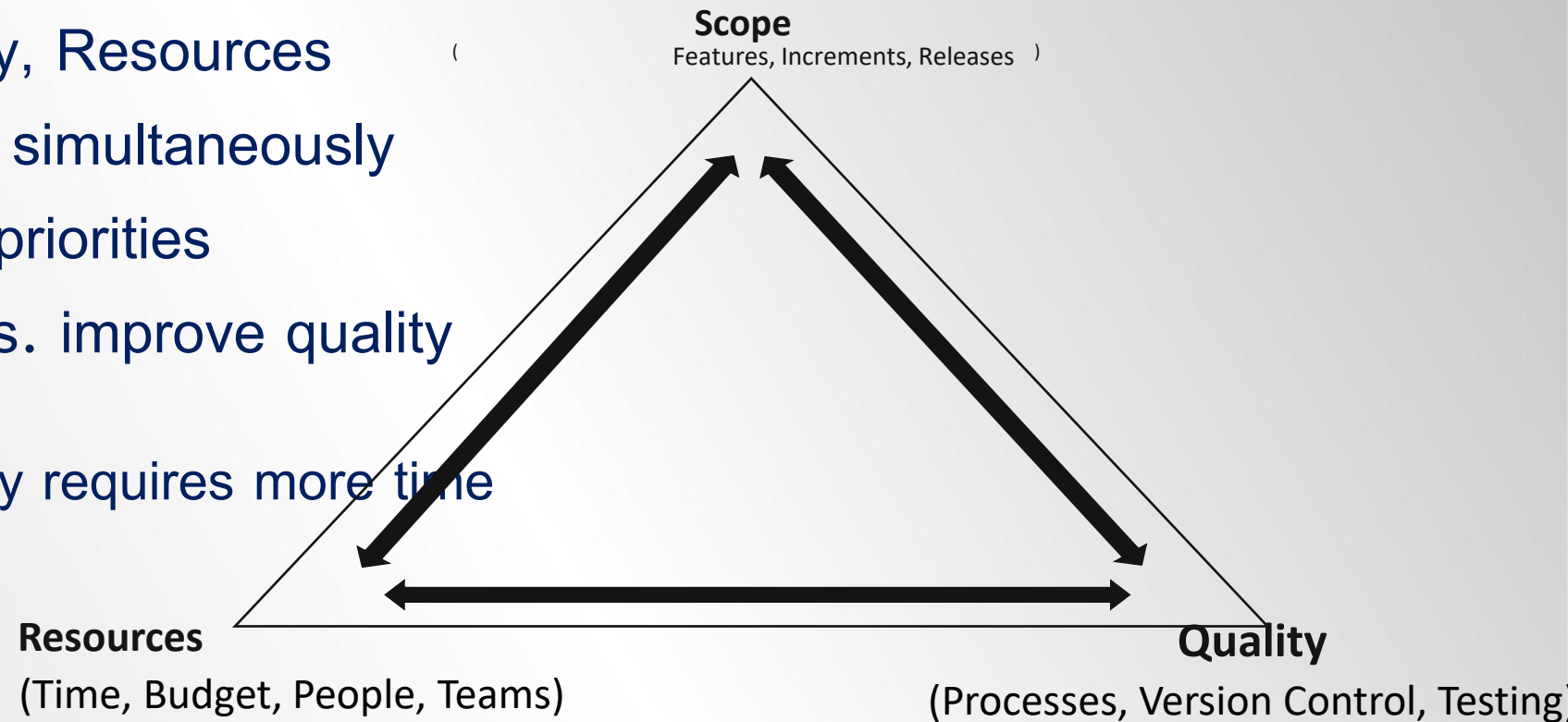
- **Iterative cycles (Plan → Design → Build → Test)**
- **Includes:**
  - Agile ceremonies (Chap. 13)
  - Lean principles (Chap. 14)
  - Version control (Chap. 15)
  - Test automation (Chap. 16)

*Notes: Reinforce that process isn't bureaucracy—it's a rhythm that builds working software reliably.*



# Engineering Trade-offs – The Software Triangle

- Triangle = Scope, Quality, Resources
- Can't maximize all three simultaneously
- Decisions must balance priorities
- Example: Add features vs. improve quality within limited time
  - Improving quality usually requires more time or fewer features





# Tabby Cat Project – A Case Study

A live example across chapters:

- Getting started (Chap. 6)
- Feature development (Chap. 12)
- Tooling and processes (Chap. 17)
- Tracks developer activity via code commits
- It is based on Feature-driven agile development.
  - Delivery of complete functionality, feature by feature



# Create Your Livelihood

Skills apply to:

- Local work
- Global freelancing
- Tech entrepreneurship

Software has power to:

- Improve health
- Strengthen communities
- Support diversity and inclusion



*Software Engineering (3) Course (SWE521): Agile  
Software Engineering Skills*

# **AGILE SOFTWARE ENGINEERING SKILLS**

**END OF LECTURE 01**

**Introduction and Principles**

**NEXT**



**Lecture 02: Self-Organizing Teams**